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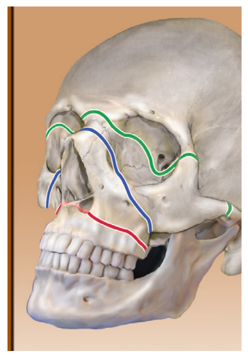

Infectious Diseases in the Athlete

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Athletic activities and the competitive sports environment present the athlete with unique exposures and risks for infectious diseases. Infectious diseases are responsible for up to 50% of visits in high school and college training rooms. Infections in athletes pose a significant public health concern because of a high frequency of close social and physical contact with teammates, coaches, support staff, and spectators, especially in organized sports at the high school, college, and professional levels. In most cases, the morbidity associated with infectious diseases is mild in this generally healthy population. However, in some rare cases, significant morbidity or even death may occur. Moreover, infectious diseases can disrupt the performance of an individual athlete or a team in competition.[1]

EPIDEMIOLOGY

In sports, the three general modes of transmission of infection are (1) direct person-to-person contact (i.e. through skin), (2) indirect contact (e.g. respiratory, blood-borne, or fecal-oral), and (3) via a common source (e.g. shared water coolers, athletic equipment, locker rooms, whirlpools, swimming pools, and contaminated freshwater venues).[2] Athletes may be more susceptible to infection than the general population for several reasons. They come into close physical contact with each other and may share personal items and equipment. Furthermore, athletes often live in dormitories or hotel rooms while traveling and come into close contact with one another in these settings. Several studies suggest that athletes may be more likely than the general population to engage in risky behaviors.[3] For example, fewer athletes practice safe sex, and athletes are more likely to use illicit drugs, alcohol, and injectable substances such as steroids and hormones, placing them at higher risk for intravenous needle exposures.[3] Last, the growing popularity of adventure sports and ecotourism places participants at risk for exotic illnesses.

Turbeville and colleagues[2] performed a systematic review of the literature characterizing infectious disease outbreaks among athletes that occurred between 1966 and 2005. The skin was the most common site of infection, representing 56% of reported infectious disease outbreaks. Direct, person-to-person (skin-to-skin) contact was the most common mode of transmission. As might be expected, the majority of outbreaks occurred in high-contact sports such as football (34%), wrestling (32%), and rugby (17%). Outbreaks were also reported in soccer, adventure races, swimming, triathlon, track and field, gymnastics, basketball, and fencing. The most common pathogens are as follows: herpes simplex virus (HSV) (22%), *Staphylococcus aureus* (22%), enteroviruses (19%), tinea (14%), *Streptococcus pyogenes* (7%), hepatitis A and B viruses (7%), measles virus (5%), leptospira (3%), and *Neisseria meningitides* (3%). Norwalk virus, rickettsia, chlamydia, and pseudomonas infections were also reported. Similarly, in a review of infectious disease outbreaks in competitive sports from 2005 to 2010, skin and soft tissue were the most common sites of infection (71%). In this study, methicillin-resistant *S. aureus* (MRSA) (33%) and tinea (29%) were the most common pathogens.[4] It is important to note that nonoutbreak infections also occur and that infections common in the general population are also common in athletes.

EXERCISE AND THE IMMUNE SYSTEM

Physical exercise is known to have important effects on the human immune system that may affect an athlete's risk of infection. The human immune system consists of the innate immune system and the adaptive or acquired immune system. The innate immune system is set up to fight infection regardless of whether a person has been previously exposed. The innate system comprises physical and functional barriers including the skin, mucous membranes, nasal hairs, areas of extreme temperature and pH, and debris removal systems such as the gastrointestinal tract and the mucociliary elevator.[5] The innate immune system also fights infection through natural killer (NK) cells, phagocytes, and proteins such as tumor necrosis factor, cytokines, and the complement system.[5] The acquired immune system is activated by exposure to specific antigens and provides long-lasting protection against anything it has previously encountered, but it does not immediately begin to fight new pathogens. It comprises T and B lymphocytes and their products, immunoglobulins (Ig), and cytokines. The acquired immune system produces IgM about 7 days after exposure to a new pathogen and IgG about 7 days later.[5] Secretory IgA resides in mucous membranes and acts as a first line of defense against infection.

The relationship between the intensity and duration of exercise and the incidence of upper respiratory tract infections (URTIs) tends to follow a J-shaped curve, as illustrated in Fig. 17.1.[6] It is thought that other infectious diseases may follow a similar pattern. Persons who exercise at moderate intensity have the lowest incidence of URTIs. Sedentary persons have a higher incidence of URTIs, whereas extreme exercisers have the highest

Abstract

Athletic activities and the competitive sports environment present the athlete with unique exposures and risks for infectious diseases. Infectious diseases are responsible for up to 50% of visits in high school and college training rooms. Infections in athletes pose a significant public health concern because of a high frequency of close social and physical contact with teammates, coaches, support staff, and spectators, especially in organized sports at the high school, college, and professional levels. In most cases, the morbidity associated with infectious diseases is mild in this generally healthy population. However, in some rare cases, significant morbidity or even death may occur. Moreover, infectious diseases can disrupt the performance of an individual athlete or a team in competition.

Keywords

infections
infectious diseases
athlete
athletes
sports

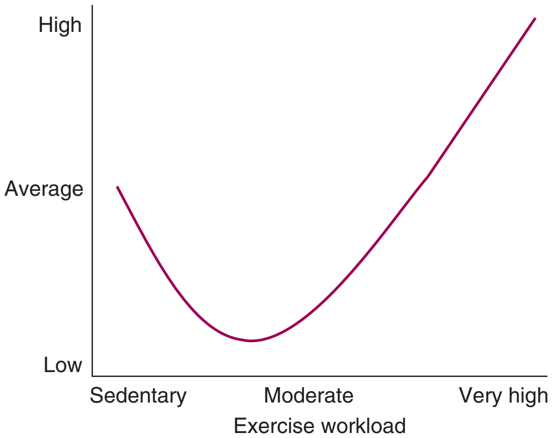

Fig. 17.1 The relationship between exercise duration/intensity and incidence of upper respiratory tract infections/infection risk (J curve). (Modified from Nieman DC. Current perspective on exercise immunology. *Curr Sports Med Rep.* 2003;82:239–242.)

risk. Moderate exercise, defined as exercise for 5 to 60 minutes within a range of 40% to 60% of the maximum heart rate, enhances the immune system by increasing neutrophil and NK counts as well as salivary IgA concentrations.[5] On the other hand, intense exercise, defined as 5 to 60 minutes of exercise at 70% to 80% of maximal heart rate, and prolonged exercise, that is, lasting more than 60 minutes, has detrimental effects on the immune system. An increased requirement for oxygen necessitates transition from nose breathing to mouth breathing during intense exercise.[5] In addition to bypassing the nasal hairs and turbulent flow that normally protects the lungs from exposure to pathogens, inhaling larger volumes of colder and drier hair also thickens mucus and disrupts the mucociliary elevator.[5] Thus greater quantities of foreign particles reach the lungs, and the body's natural defenses are impaired in their ability to clear them. With intense exercise, NK cell, lymphocyte, and neutrophil cell numbers fall, B-cell function decreases, and secretory IgA and serum IgG1, IgG2, and IgE concentrations decline.[5] Levels of cortisol, prolactin, adrenaline, and growth hormone increase, thereby impairing cell immunity.[5] A decrease in salivary lactoferrin and lysozyme concentrations impairs mucosal immunity.[5] All of these factors contribute to a relative immunosuppression that may increase the risk of infection after intense exercise.

PREVENTION

Preventive efforts should focus on primary prevention—that is, avoiding infection before it occurs. Primary prevention includes immunization and good personal hygiene. Secondary prevention consists of infection control measures aimed at preventing the spread of disease to others or recurrence in the source patient, which may include isolation of infected persons or, in some cases, postexposure prophylaxis.

Immunization

Active immunization stimulates an immune response to protect against future exposure to an infectious organism by administering all or part of a microorganism or a modified product of the microorganism such as antigens or proteins. Most vaccines are more than 90% effective, but they are not guaranteed to provide immune protection.[3] Before participation in organized sports, a physician should verify that the athlete has received appropriate immunizations and administer "catch-up immunization" if necessary. Athletes traveling abroad should consider the diseases endemic in the geographic area to which they are traveling. Ideally, immunizations should be planned at least 4 months in advance to ensure adequate time for administration.

Hygiene

Good personal hygiene can help reduce the transmission of infectious agents among athletes. The following general measures are advisable[3]:

- Use universal body fluid precautions; wash hands frequently, and always use disposable gloves when coming into contact with bodily fluids, the oral cavity, or wounds.
- Avoid sharing personal items (e.g., soap, towels, and water bottles), food, and water.
- Routinely clean shared equipment; bleach diluted with tap water in a 1 : 10 ratio is an effective cleanser.
- Any athlete with skin lesions that are weeping or leaking blood or serous fluid should be excluded from play until the area has dried and can be securely covered with occlusive bandages or dressings.
- Athletes should be instructed to report all illnesses or skin lesions promptly.

Return-to-Play Guidelines

Appropriate and timely guidelines are pivotal for prevention. The decision about whether an athlete may return to play (RTP) after experiencing an infectious illness is based on a number of factors, including the judgment of physicians, coaches, and individual athletes. The first consideration is the effect of athletic activity on the health and safety of the individual athlete. Another important consideration is preventing the transmission of infectious diseases to teammates, support staff, coaches, and spectators. Frequent close contacts during practice, in locker rooms, and in shared living spaces as well as in sharing equipment and facilities create abundant opportunities for spreading and acquiring illness.

Exercise may prolong or intensify the disease course or cause dangerous complications. Fever inhibits fluid and temperature regulation while also impairing coordination, concentration, muscle strength, aerobic power, and endurance.[5] Viral illnesses contribute to tissue wasting, muscle catabolism, and negative nitrogen balance.[5] Drugs used to treat infectious diseases may also have considerable effects on the athlete. For example, quinolone antibiotics carry a risk of tendon rupture. The use of compounds that contain ephedrine, which is banned by many sports organizations, may lead to the disqualification of an athlete. Finally, illness may limit an athlete's performance in competitive sports. Pain, discomfort, and other symptoms of infection can be distracting during play.

A good rule of thumb traditionally used in sports medicine is Eichner's "neck check."[7] Athletes should not work out while experiencing systemic symptoms (e.g., fever or myalgias) or symptoms below the neck (such as a hacking cough, vomiting, or diarrhea). If symptoms are limited to above the neck, such as a runny nose, watery eyes, or sore throat, the athlete can attempt a "test drive." The athlete exercises at "half speed," and if he or she feels better after 10 minutes, it should be safe to complete the workout at a tolerated intensity. Where specific RTP guidelines exist, they are discussed in their corresponding sections.

SKIN AND SOFT TISSUE INFECTIONS

Skin and soft tissue infections comprise most common infectious disease outbreaks in athletes, representing anywhere from 56% to 71% of outbreaks.[2,4] Participants in contact sports such as wrestling, football, and rugby are at highest risk for acquiring such an infection. Frequent skin trauma, moist environments, and direct contact with equipment and other players are all factors that make athletes particularly vulnerable to these infections. The most common causative agents are *S. aureus* and HSV, each constituting 22% of reported infectious disease outbreaks.[2] During the past decade, MRSA has emerged as a common cause of superficial skin infections in the community and has been known to affect a significant number of athletes at the high school and collegiate level. Although most skin infections resolve without complication, they are highly contagious and can result in significant loss of playing time; thus they warrant prompt recognition and treatment. The Centers for Disease Control and Prevention[8] and the American College of Sports Medicine[9] recommend the following preventive measures:

- Practice good personal hygiene: minimize contact, wash hands frequently, shower with soap and hot water after all practices and competitions, and wear sandals in public showering facilities.
- Avoid sharing items that come in contact with the skin (e.g., razors, clothing, linens, and towels).
- Discourage body shaving, which increases the risk of trauma and predisposes to infection.
- Cover wounds (including abrasions, blisters, and lacerations) until they have healed. When wounds cannot be properly covered (by a securely attached bandage that will contain all drainage and remain intact throughout the activity), athletes should be excluded from play.
- Immediately inform coaches of any skin lesions.
- Athletes with active infections or open wounds should avoid common-use water facilities such as swimming pools.
- Thoroughly clean high-touch surfaces (i.e., surfaces that come in frequent contact with people's bare skin each day), including counters, doorknobs, bathtubs, and toilet seats.

When an athlete is diagnosed with a skin infection, care should be taken to appropriately document the diagnosis and treatment to ensure adequate treatment and guarantee meeting return-to-play guidelines. Documentation should include the diagnosis, culture results, date and time of initiation of therapy, and exact names of medications used.[10]

Bacterial Skin Infections

Definition

Superficial bacterial skin infections are most often caused by *Streptococcus* or *Staphylococcus* species.[11] The prevalence of MRSA has risen dramatically in the community during the past decade. MRSA is a type of staphylococcal bacteria that is resistant to traditional penicillin-based antibiotics and has historically affected patients in health care settings.

Epidemiology

Open or broken skin is the biggest risk factor for bacterial skin infection. Close skin-to-skin contact, contact with contaminated items and surfaces, crowded living conditions, and poor hygiene also place a person at risk for bacterial skin infection.[12] Whereas 25% to 30% of people are colonized in the nose with *Staphylococcus,* fewer than 2% are colonized with MRSA.[13] *S. aureus* and HSV are the most common causes of infectious disease outbreaks in athletes, with each responsible for 22% of outbreaks.[2]

Pathobiology

Bacterial skin infections have a variety of clinical presentations. *S. aureus* and MRSA commonly present as folliculitis, furuncles or "boils" (abscessed hair follicles), carbuncles (coalesced masses of furuncles), and abscesses. Impetigo describes a weepy skin lesion with a honey-colored crust. Erysipelas (a superficial infection of the skin with well-demarcated borders) and cellulitis (subcutaneous involvement with possible systemic symptoms) are most commonly caused by group A *Streptococcus* and *Staphylococcus* species in the athletic population.[12] More virulent strains of *Staphylococcus*, including MRSA, can cause osteomyelitis, sepsis, toxic shock syndrome, and necrotizing pneumonia.[9]

Diagnosis

The hallmarks of a bacterial skin and soft tissue infection include redness, swelling, warmth, and pain/tenderness. MRSA should be considered in the differential diagnosis of any such skin lesions. Cellulitis and abscesses are both considered to be types of superficial wound infections. Cellulitis demonstrates features of both skin erythema and increased warmth and can be more diffuse in nature (Fig. 17.2). An abscess, on the other hand, involves a localized collection of purulent material (Fig. 17.3).[1] MRSA

Fig. 17.2 Orthopedic cellulitis. (Courtesy University of Virginia Department of Dermatology, Kenneth Greer, MD, Chairman.)

Fig. 17.3 An abscess. (Courtesy University of Virginia Department of Dermatology, Kenneth Greer, MD, Chairman.)

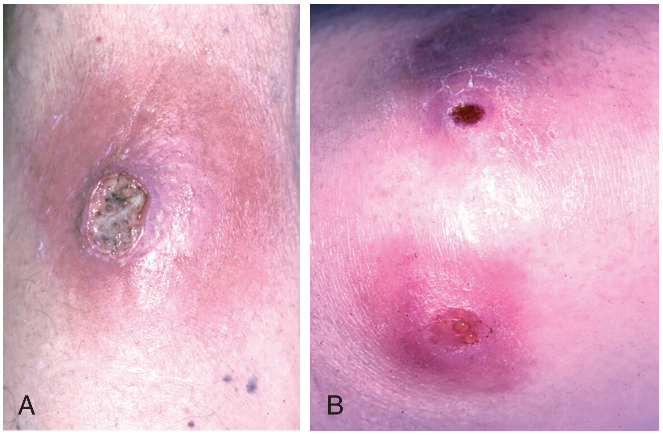

Fig. 17.4 (A and B) Examples of community-acquired methicillin-resistant *Staphylococcus aureus* (MRSA) infections of the skin. Lesions that have signs of necrosis, resemble spider bites, or have pain and erythema out of proportion to visual inspection are highly suggestive of community-acquired MRSA. (Courtesy University of Virginia Department of Dermatology, Kenneth Greer, MD, Chairman.)

infections are commonly mistaken for spider bites. On physical examination, lesions suspicious for MRSA are purulent, exhibiting fluctuance (i.e., a palpable, movable, compressible fluid-filled cavity), a yellow or white center, and a central point or head (Fig. 17.4A and B).[1] They are commonly found at sites of visible skin trauma and areas of the body covered by hair. Nonpurulent cellulitis or erysipelas strongly suggests *Streptococcus* as the causative agent. Bacterial skin infections may resemble insect bites, trauma, superficial burns, contact dermatitis, acne, tinea, dermatophytes, or HSV.[14] A gram stain may elucidate the diagnosis in uncertain cases. A culture of the skin lesion may be useful in cases of recurrent or persistent infection, antibiotic failure, or advanced or aggressive infections.

Treatment and Prevention

In 2011, the Infectious Diseases Society of America released evidence-based guidelines for the empiric treatment of bacterial skin infections based on clinical features.[15] Simple infections presenting as impetigo, simple abscesses, furuncles, and carbuncles may resolve by applying moist heat and/or applying mupirocin topically twice daily for 10 days. Incision and drainage is the mainstay therapy for simple infections that do not resolve when treated with moist heat and mupirocin. Drainage fluid should be sent for culture and susceptibility testing to direct antibiotic therapy. In addition to incision and drainage, empiric antimicrobial therapy with coverage for MRSA should be considered if the lesion is severe or extensive (e.g., involving multiple sites, associated with cellulitis, or having signs and symptoms of systemic illness), located in an area that is difficult to drain (such as the face, hand, and genitalia), or fails to respond to incision and drainage alone after 48 hours, or if the patient is immunosuppressed or at the extremes of ages. Five to 10 days of therapy is recommended. If cellulitis is present without evidence of purulence or abscess, the causative organism is more likely to be *Streptococcus* and can be treated with a β-lactam antibiotic (e.g., first-generation cephalosporin). If it fails to improve or the patient experiences systemic symptoms, treating physicians should strongly consider the possibility of community-associated MRSA. The treatment of purulent cellulitis warrants empiric treatment for MRSA with trimethoprim-sulfamethoxazole, doxycycline, clindamycin, linezolid, or minocycline.

No clear evidence exists for the effectiveness of decolonization therapy for recurrent MRSA skin and soft tissue infections. However, decolonization may be considered if (1) the patient experiences recurrent skin and soft tissue infections despite optimizing hygiene and wound care or (2) ongoing transmission is occurring among close contacts despite optimization of hygiene and wound care. Decolonization methods include use of nasal mupirocin, skin antiseptic solution (e.g., chlorhexidine), or dilute bleach baths. Oral antimicrobial therapy is recommended only for the treatment of active infection. An oral agent in combination with rifampin may be considered for decolonization if infections recur despite use of the aforementioned methods.

Return-to-Play Guidelines

RTP guidelines for all bacterial skin infections (e.g., furuncles, carbuncles, folliculitis, impetigo, cellulitis, erysipelas, staphylococcal disease, and community-acquired MRSA) are grouped together. As published in the National Collegiate Athletic Association (NCAA) guidelines for wrestling, the following criteria apply before returning to play[10]:

- The athlete must have no new skin lesions for 48 hours before participating in a tournament or practice.
- The athlete must have completed 72 hours of oral antibiotic therapy (the National Federation of High Schools requires oral antibiotic therapy for 48 hours).
- The athlete must have no moist, exudative, or draining lesions while participating in a tournament or practice.

Although no specific guidelines are outlined for other sports, the guidelines for wrestling should be followed for other contact sports (such as football and rugby), sports with shared equipment or facility use (such as gymnastics or aquatic sports), and noncontact sports on a case-by-case basis. All dry lesions should be covered during play.

Herpes Simplex Virus

Definition

HSV can cause primary or recurrent infections and is highly contagious. HSV infections are exceedingly common in the general population and sufficiently prevalent in the athletic population that special terms have been coined to describe outbreaks in sports medicine: *Herpes gladiatorum* was originally used to describe HSV infection in wrestlers, and *scrumpox* to describe HSV in rugby players.[16,17]

Epidemiology

HSV is responsible for 22% of infectious disease outbreaks in athletes, making it and HSV the most common pathogens encountered. One survey reported that the annual incidence of HSV lesions was 7.6% among college wrestlers and 2.6% among high school wrestlers.[8] Transmission can occur by direct skin-to-skin contact or bodily fluids including saliva, semen, and vaginal secretions. It is estimated that the likelihood of contracting herpes when sparring with an infected partner during an active outbreak is 32.7%.[8]

Pathobiology

HSV-1 is the most common cause of herpes labialis (affecting the lips), and HSV-2 is the most common cause of urogenital herpes. Primary infection occurs after an incubation period of 2 to 20 days and may be accompanied by systemic symptoms including fever, adenopathy, malaise, myalgias, and headache. HSV remains latent in the neural ganglia, and reactivation may occur in the setting of reexposure, autoinoculation, physical or emotional stress, poor nutrition, ultraviolet radiation, fever, coexisting infection, or immunosuppression. Reactivation of skin lesions is typically preceded by a prodromal phase of neuralgia, tingling, or burning sensations.[16] Systemic symptoms are typically absent in reactivation. Lesions are most commonly present on the lips, head, extremities, and trunk but can also affect the eyes.

Diagnosis

Diagnosis is typically made on the basis of the characteristic clinical appearance of the lesions. Herpetic lesions form a cluster of vesicles on an erythematous base. Vesicles may ulcerate and leave a shallow painful ulcer with surrounding erythema. Lesions subsequently crust or scab while healing, and complete resolution may take up to 2 to 3 weeks. Laboratory testing can be performed to confirm questionable diagnoses but is not required. Viral isolation from tissue culture is the test of choice.[9] A Tzanck test of fluid from a vesicle reveals characteristic multinucleated giant cells.

Treatment

Oral systemic antiviral medication for 5 days is the standard treatment for HSV outbreaks to reduce the duration of the outbreak and time to complete lesion healing. Medications are effective if taken within the first 48 hours of the appearance of any lesion. Acyclovir, valacyclovir, and famciclovir are commonly used. Athletes with a history of recurrent herpes labialis or herpes gladiatorum should be considered for season-long suppressive therapy. Ocular herpes requires an urgent ophthalmologic referral.

Return-to-Play Guidelines

Athletes engaged in contact sports should refrain from playing until all lesions are dry with a firm adherent crust and until the athletes have been taking an appropriate dose of systemic antiviral therapy for at least 120 hours. The NCAA guidelines for RTP for wrestlers after HSV infection are summarized in Table 17.1.[10] Although no official guidelines are available for other sports, these same guidelines can be used to guide RTP for all other contact sports.

Fungal Skin Infections

Definition

Superficial fungal skin infections are caused by dermatophytes, and nomenclature is based on the location of lesions on the body. Tinea capitis refers to infection of the skin and hair on

TABLE 17.1 National Collegiate Athletic Association Guidelines for Return to Play After Herpes Simplex Virus Infection

Nature of Infection	Guidelines
Primary	Free of systemic symptoms (e.g., fever and malaise) No new blisters for 72 h before practice or tournament No moist lesions; must be dried and surrounded by a firm adherent crust Must have been taking an appropriate dose of a systemic antiviral therapy for at least 120 h before and at the time of practice or play in a tournament Active herpetic infections shall not be covered to allow participation
Recurrent	No moist lesions; must be dried and surrounded by a firm adherent crust Must have been taking an appropriate dose of systemic antiviral therapy for at least 120 h before and at the time of practice or play in a tournament Active herpetic infections shall not be covered to allow participation
Questionable cases	Tzanck preparation and/or herpes simplex virus antigen assay Participation deferred until Tzanck prep and/or herpes simplex virus antigen assay results complete

Modified from National Collegiate Athletic Association Rules Committee. Skin infections in wrestling (Appendix C). In *2017–2018 and 2018–2019 NCAA Wrestling Rules Book*. Indianapolis, IN: National Collegiate Athletic Association; 2017.

the scalp. Tinea corporis, also known as ringworm, refers to infections on the body. Tinea cruris, or jock itch, is the term used to describe groin infections. Foot infections are called tinea pedis or athlete's foot. Tinea gladiatorum describes fungal infections of the skin or scalp in athletes.

Epidemiology

Fungal infections affect between 10% and 20% of the population worldwide, with tinea pedis being the most common clinical manifestation; 70% of adults experience tinea pedis in their lifetimes.[16] Tinea pedis is common in athletes, with an infection rate of about 35%; it is especially common in swimming pool users and marathon runners.[16]

Pathobiology

The causative agents of fungal skin infections are dermatophytes. The major genera of dermatophytes are *Trichophyton, Microsporum,* and *Epidermophyton*. Dermatophytes are transmitted by direct person-to-person contact, animal-to-human contact, contact with fomites, or directly from the soil. Dermatophytes infect the stratum corneum layer of the skin. The host responds by increasing proliferation of the basal cell layer, resulting in epidermal thickening and scale formation. The characteristic tinea corporis lesion is a singular, well-defined, erythematous plaque with an expanding red, raised ring and a central clearing, often accompanied by flaking and pruritus. Tinea corporis in athletes most commonly affects the head, neck, trunk, and upper extremities. Tinea capitis is characterized by an annular patch of hair loss and a gray hyperkeratotic plaque. Tinea cruris affects the pubic area, inguinal folds, and medial thighs. Tinea pedis can present as the interdigital type, with red, weeping, macerated skin and fissures in the web space between toes; the moccasin type, with plaques on the sole and sides of the foot; or the bullous type, with vesicles or bullae filled with clear fluid.

Diagnosis

Dermatophyte infection may be diagnosed clinically based on the characteristic appearance of the skin lesions. Questionable cases may be confirmed by direct microscopy of a potassium hydroxide (KOH) preparation demonstrating septated hyphae. A fungal culture using Sabouraud dextrose agar may be performed if lesions are suspicious but the KOH preparation is negative. Tinea capitis can be distinguished from other causes of localized alopecia by its characteristic gray hyperkeratotic plaque. Tinea corporis and tinea cruris may be distinguished from impetigo, psoriasis, lichen planus, seborrheic dermatitis, pityriasis rosea, and secondary or tertiary syphilis with use of a KOH preparation or fungal culture. Erythrasma, a *Corynebacterium* infection, may resemble tinea cruris, but examination under a Wood light reveals a coral-red color.

Treatment and Prevention

Topical or oral medications may be used to treat cutaneous fungal infections. For the general population, topical therapy is the first-line treatment for tinea corporis and tinea cruris. Oral medications can have significant adverse effects and are thus reserved for extensive or disabling disease, patients for whom topical therapy has failed, and patients who are immunosuppressed. However, tinea capitis requires treatment with an oral agent. When possible, wrestlers and participants in contact sports should also be treated with oral fungicidal medications.[16] Topical agents include imidazoles, allylamines, and napthiomates. The standard duration of treatment is 2 to 4 weeks, and common regimens include terbinafine 1% cream one to two times daily, ketoconazole 2% cream once daily, or Clotrimazole 1% one to two times daily. Oral agents include fungicidal drugs such as allylamines and fungistatic drugs such as imidazoles and griseofulvin. Fungicidal drugs are often preferred because they require shorter courses of therapy.[16] Showering daily after practice or play, drying thoroughly, and wearing cotton socks and underwear can help reduce the incidence of fungal rashes.

Return-to-Play Guidelines

As with other skin infections, athletes with active tinea lesions should refrain from participating in contact sports until the infection has cleared and been adequately treated to avoid transmission to other athletes. The affected athlete should have completed a minimum of 72 hours of topical therapy for skin lesions and a minimum of 2 weeks of systemic antifungal therapy for scalp lesions.[10] Resolution of treated lesions can be evaluated by KOH preparation or review of the therapeutic regimen. All lesions should be covered with a gas-permeable dressing. If lesions are extensive and cannot be adequately covered, the athlete should refrain from play.

UPPER RESPIRATORY TRACT INFECTIONS

Epidemiology

A URTI, or the common cold, is the most common acute illness in the general population. It affects every healthy adult one to six times each year, especially during the fall and winter seasons, and thus is also the most common infection seen among athletes. Respiratory tract infections were found to be among the most common causes of cough in the athletic population.[18] Transmission occurs through (1) direct contact, when secretions are transferred from hand to hand, then from hand to mucus membranes of the nose or eyes, or (2) respiratory transmission via small-particle aerosols and large-particle droplets. Athletes have an elevated risk of exposure because of the high frequency of close contact while practicing and traveling together.

Pathobiology

The vast majority of URTIs have a viral etiology: rhinoviruses (40%), coronaviruses (20%), respiratory syncytial virus (10%), influenza virus, parainfluenza virus, and adenovirus.[19] Two to 3 days after exposure a patient typically experiences rhinorrhea, cough, and fever. URTIs are generally self-limiting and last 5 to 14 days. Complications may include acute sinusitis, lower respiratory tract infection, otitis media, and asthma exacerbation. URTIs are a common trigger for asthma attacks and are estimated to be responsible for 40% of asthma attacks.[19]

Inflammation of the paranasal sinuses, especially maxillary and frontal sinuses, are caused by the same viruses that cause URTIs, and approximately 2.5% of adult patients experience acute bacterial sinusitis as a complication after a URTI.[5] Acute bacterial sinusitis is most commonly caused by *Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis, S. aureus,* and anaerobes. Athletes who participate in swimming, diving, water polo, and surfing seem to have a higher incidence of sinusitis.

Diagnosis

Laboratory testing is not routinely needed for URTIs; the diagnosis is clinical and treatment is based on symptoms. Acute bacterial sinusitis should be suspected in patients whose URTI symptoms are accompanied by purulent nasal discharge and maxillary or facial pain and who demonstrate a poor response to decongestants or abnormal transillumination.[19] Sinus aspirate culture is the gold standard for diagnosing sinusitis, but it is rarely performed because of its invasive nature. Imaging studies such as a computed tomography (CT) scan or magnetic resonance imaging (MRI) may be used if disease persists despite optimal therapy.

Treatment and Prevention

URTIs are treated symptomatically with oral decongestants and antihistamines or nasal decongestant sprays, analgesics, and antipyretic agents. The use of compounds that contain ephedrine, including many oral decongestants, is banned by many sports organizations and may lead to disqualification of an athlete. Athletes should take care to remain adequately hydrated. If the URTI is caused by an influenza virus, a neuraminidase inhibitor such as zanamivir or oseltamivir may decrease the severity and duration of symptoms if it is initiated within 48 hours of exposure.[3] Antibiotics are not beneficial for simple URTIs but may be useful for acute bacterial sinusitis if symptoms worsen over 5 to 7 days or fail to improve after 10 days. Empiric treatment for acute bacterial sinusitis is amoxicillin for 10 to 14 days.[19] Influenza vaccine has an efficacy of 70% to 90%; thus athletes without contraindications should be vaccinated annually.[5]

Return-to-Play Guidelines

In the case of a simple URTI, RTP generally depends on the athlete. It can be expected that athletic performance may be adversely affected during illness, but exercise does not seem to alter the course of the illness. Sedentary subjects who exercised after getting a URTI did not experience an effect on symptoms or duration of illness.[20,21] The "neck check" principle applies; as long as the patient is afebrile and symptoms described are "above the neck," an athlete may continue physical activity as tolerated.

Infectious Mononucleosis

Definition

Infectious mononucleosis (IM), a common medical condition caused by Epstein-Barr virus (EBV), warrants special mention for two reasons: (1) epidemiologically, it is most clinically significant in the adolescent and young adult age group, which is the primary age group engaged in athletic activity, and (2) the most feared complication of IM, splenic rupture, raises important questions for sports medicine physicians making decisions for suitability of an athlete to RTP.

Epidemiology

The prevalence of IM in the general population is 45 cases for every 100,000 people.[22] Antibodies to EBV eventually develop in approximately 95% of adults, indicating prior infection.[22] The peak incidence of IM occurs in 15- to 25-year-olds, and this age group is also most likely to experience acute symptomatic infection. Symptoms rarely develop in adults older than 35 years or in children younger than 15 years.

Pathobiology

EBV is a DNA herpesvirus that is transmitted via oropharyngeal secretions; it is thus referred to as the "kissing disease." It may be acquired through sharing drinks or eating utensils and through aerosolized secretions from sneezing and coughing. EBV enters squamous epithelial cells of the oropharynx and has also been isolated from epithelial cells of the cervix and semen. It is often difficult to identify the source because EBV has a very long incubation period of 30 to 50 days before the onset of symptoms. The classic triad of IM is fever, sore throat, and lymphadenopathy. Patients often experience a prodrome of malaise, headache, and high fever that can last up to 3 weeks. Splenomegaly occurs in 50% to 100% of patients. A rash on the trunk and upper arms occurs in 10% to 40% of patients and is more common in persons treated with ampicillin or amoxicillin.[23]

Whereas most cases of IM are moderate in severity and resolve without complication, complications have the potential to be severe. Reported but rare complications include Guillain-Barré syndrome, meningitis, neuritis, hemolytic uremic syndrome, disseminated intravascular coagulation, and aplastic anemia. In the athletic population, three complications are of particular concern:

1. Severe tonsillar enlargement leading to acute respiratory compromise
2. Prolonged fatigue limiting return to the preillness level of activity for up to 3 months
3. Splenic rupture, which, although extremely rare, usually occurs within the first 3 weeks of symptomatic illness[22]

It is thought that lymphocytic infiltration of the spleen disrupts its normal anatomy, weakening supporting structures and increasing splenic fragility. Rupture can be caused by trauma in the setting of a sudden increase in portal venous pressure from a Valsalva maneuver or compression from external trauma. The occurrence of spontaneous rupture in the absence of trauma has been reported in only a handful of cases.[24]

Diagnosis

Clinical assessment remains the predominant method of diagnosing IM. The differential diagnosis should include viral or bacterial pharyngitis, *Streptococcus, Gonococcus,* human immunodeficiency virus (HIV), and cytomegalovirus. It is particularly important to test for *Streptococcus*, with which an estimated 30% of patients with IM are coinfected.[22] Untreated group A, β-hemolytic streptococcal infections may result in glomerulonephritis or rheumatic fever.

On physical examination, fever is often high and can be greater than 104°F. Exudative tonsillar pharyngitis is present and may be accompanied by palatal petechiae. Lymphadenopathy is typically painful and most commonly affects the posterior cervical chain. Axillary and inguinal involvement may help to differentiate IM from other forms of viral or bacterial pharyngitis. Laboratory findings of leukocytosis in the range of 12,000 to 18,000 white blood cells with greater than 50% lymphocytes and at least 10% atypical lymphocytes on a peripheral blood smear are highly suggestive of IM. The monospot test for heterophile antibodies is specific but not sensitive, especially early on in the disease course. The false-negative rate is 25% in the first week but drops to 5% in the third week. In children younger than 10 years, the monospot test detects fewer than 50% of EBV infections.[22] In a small minority of patients, a seropositivity for heterophile antibodies will never develop.[9] When the monospot test is negative and IM is strongly suspected, EBV titers can be obtained. Specific antibody tests for EBV include viral capsid antigen (VCA), IgG, and IgM antibodies, along with antibody to Epstein-Barr nuclear antigen. VCA IgM appears early in infection and disappears after 4 to 6 weeks. VCA IgG peaks at 2 to 4 weeks after onset and then declines slightly and persists for life. Epstein-Barr nuclear antigen antibody appears after 2 to 4 months and persists for life.[22]

Splenomegaly is almost universal, but no gold standard exists for its diagnosis. The utility of bedside examination to detect splenomegaly is limited. Both palpation and percussion are recommended, but they have a sensitivity of 46% and a specificity of 97%.[22] Physical examination may further be limited by the rigid abdominal musculature of athletes. Few data exist to support the use of diagnostic imaging to evaluate for splenic size. When evaluating for splenomegaly, serial measurement by ultrasound is preferred rather than CT because of its lower cost and the avoidance of radiation exposure. Although the spleen can be accurately sized, spleen size varies greatly between persons, and no normative data are available. Without knowing a baseline spleen size, one cannot determine whether a patient has splenomegaly, but serial studies may be helpful. If splenic injury is suspected, CT or MRI is preferable to ultrasound to acquire higher-resolution images.

Treatment and Prevention

IM is generally self-resolving and thus the mainstay of treatment is symptomatic relief and supportive care: fever control, analgesia for pharyngitis, rest, and hydration. Antiviral medications have not been shown to diminish the severity or duration of symptoms.[9] All patients should be evaluated for coexisting streptococcal pharyngitis. If present, it should be treated with antibiotics other than amoxicillin and ampicillin, which may provoke rashes in patients with IM. Oral corticosteroids may be helpful if the course is complicated by hepatitis, myocarditis, hemolytic uremic syndrome, neurologic complications, or airway obstruction. Patients should be advised to avoid consumption of excessive alcohol, acetaminophen, and other liver toxins. Athletes should be advised that EBV is present in oropharyngeal secretions. Avoiding contact with secretions through the sharing of food, eating utensils, water bottles, and kissing may decrease the risk of transmission.

Return-to-Play Guidelines

The decision about when an athlete can RTP after an IM infection is difficult because the risk of splenic rupture is not well characterized. The consensus statement by the American College of Sports Medicine states that light, noncontact activity may gradually be introduced as tolerated 3 weeks after illness onset provided that the patient is afebrile, has good energy levels, and has no other complications.[9] Although splenic rupture can occur with trauma, most ruptures are atraumatic and occur in the first 3 weeks of illness. During this time, athletes should be advised to avoid chest or abdominal trauma, significant exertion, or Valsalva activities including weightlifting and rowing.

The American College of Sports Medicine recommends avoiding contact or collision sports until 4 weeks after the onset of illness.[9] However, it is important to remember that splenic rupture has been reported to occur up to 7 weeks after the onset of illness.[22] Athletes should be aware that the risk of splenic rupture decreases with time but is never zero. Many persons advocate that the athlete should have a “normal”-sized spleen before returning to play, with the thought that risk of rupture may be highest when the spleen is enlarging. In reality, the relationship between splenomegaly and rupture is unclear. Because of the limitations previously discussed, it is not standard practice to determine splenic size using diagnostic imaging. Serial ultrasonography may be useful in cases where the patient becomes clinically asymptomatic early and is considering early RTP.[22] Prolonged fatigue may limit full performance status for as long as 3 months.[22]

Measles

Definition and Epidemiology

Measles, or rubeola, is an acute, highly contagious viral illness. Airborne measles outbreaks have been known to occur in indoor sports among gymnasts, wrestlers, basketball players, and spectators in crowded and humid gymnasiums and sports domes.[20] Measles is still endemic in many countries worldwide and results in an estimated 800,000 deaths per year. In the United States, between 1996 and 2000, the incidence of measles was less than 1 case per million.[25] Approximately 62% of cases were internationally imported, but a number of cases had an unknown source. The majority of infections occur in unvaccinated patients (46%) and in patients in whom vaccination status was unknown (27%), but 27% of cases in the year 2000 occurred in patients with a documented history of measles vaccination.[25]

Pathobiology

Measles is primarily transmitted directly via respiratory droplets traveling over short distances and less commonly via small-particle aerosols that persist in the air for a long time.[26] During the incubation period of 10 to 14 days, the measles virus replicates initially in the epithelial cells of the upper respiratory tract and then spreads to local lymphatic tissue, the blood, and many organs including the lymph nodes, skin, kidney, gastrointestinal tract, and liver. The host cellular immune response at the sites of replication produces signs and symptoms. Persons with impaired cellular immunity may have absent or delayed presentation of symptoms. The prodromal phase of measles presents with fever, cough, coryza, and conjunctivitis. Koplik spots, which are small white lesions on the buccal mucosa during the prodromal phase, are diagnostic for measles. Several days before the onset of rash, prodromal symptoms intensify. The characteristic rash is an erythematous macropapular eruption beginning on the face and then spreading to the trunk and extremities, which lasts for 3 to 5 days. Patients are infectious for several days before and after the onset of the rash.[26]

In uncomplicated cases of measles, resolution begins shortly after the appearance of the rash. However, after infection with measles, persons have depressed immunity as a result of delayed-type hypersensitivity and are thus more susceptible to secondary bacterial or viral infection.[26] Consequently approximately 40% of cases are followed by complications, including pneumonia, laryngotracheobronchitis, keratoconjunctivitis, stomatitis, and diarrhea. Rare but serious central nervous system (CNS) complications may occur 2 weeks to several years after infection, including postmeasles encephalomyelitis, measles inclusion body encephalitis, and subacute sclerosing panencephalitis.

Diagnosis

Measles should be suspected in any patient who presents with fever and a generalized rash. The differential diagnosis includes other viral exanthems such as rubella. Koplik spots are diagnostic of measles prior to the appearance of a rash. A physical examination should include investigation for secondary viral or bacterial infections. The diagnosis of acute measles can be confirmed with serology. The most common test is detection of measles virus–specific IgM in serum or oral secretions, but it may not be detectable for 4 days after rash onset and disappears within 4 to 8 weeks of infection. Levels of measles virus–specific IgG four times greater than convalescent levels may also be diagnostic of an acute infection.[26]

Treatment

The World Health Organization recommends administration of vitamin A, 200,000 international units daily for 2 consecutive days, for all patients older than 12 months to treat measles and reduce morbidity and mortality.[26] No specific antiviral agents are indicated for treatment of measles, but ribavirin, interferon-β, and other antiviral drugs are used in persons with severe infections. Antibiotics may be required if the patient is coinfected with bacteria. *S. pneumoniae* and *H. influenzae b* are common pathogens. The attenuated vaccine is the best available method of primary prevention.

Return-to-Play Guidelines

Athletes with suspected measles should be isolated from other players expediently. All cases of measles should be reported to public health authorities so that appropriate infection control measures can be administered.

LOWER RESPIRATORY TRACT INFECTIONS

Lower respiratory tract infections have two main clinical presentations: acute bronchitis and acute pneumonia.

Acute Bronchitis

Definition

Acute bronchitis is inflammation of the bronchial tree with cough lasting about 3 weeks, with or without sputum production.[5] Acute bronchitis is usually of viral origin, with the most common pathogens being influenza A and B, parainfluenza viruses, coronaviruses, rhinoviruses, and adenoviruses.

Diagnosis

Acute bronchitis is usually diagnosed clinically. Patients present with a dry cough for no more than 3 weeks. On physical examination, coarse breath sounds or wheezes are occasionally heard on auscultation. Chest radiographs are not routinely needed.

Treatment

Acute bronchitis is usually a self-limiting illness that resolves after 3 weeks; thus treatment is symptomatic. Antibiotics are rarely indicated because most cases are of viral etiology. Bronchodilators may be useful to improve respiratory flow dynamics.[5]

Return-to-Play Guidelines

No specific guidelines exist for RTP after an episode of acute bronchitis. Athletes can resume activity as long as they are afebrile, lack other systemic symptoms, and pass the “neck check” described previously.

Acute Pneumonia

Definition

Pneumonia, an infection of the lung, is viral in 30% to 50% of cases. Other pathogens include *S. pneumoniae*, *Legionella,* and *Chlamydia*.

Pathobiology

Patients present with a cough productive of purulent sputum, shortness of breath, chest pain, malaise, anorexia, fevers, and chills. Symptoms usually improve within 3 days. Fever lasts 3 days and dyspnea may last for 6 days, whereas cough and fatigue last approximately 2 weeks.[27] Of particular concern to the athletic population is that acute pneumonia may be complicated by reactive airway disease, a transient airway obstruction, and hyperresponsiveness, which may take up to 2 months to resolve and thus significantly delay RTP.[27]

Diagnosis

On physical examination, the patient may demonstrate abnormal vital signs including fever, tachycardia, tachypnea, hypoxemia, and hypotension. A chest examination may reveal dullness to percussion, crackles, rales, bronchial breath sounds, tactile fremitus, or egophony. Diagnosis of pneumonia in the context of the clinical picture requires visualization of an infiltrate on a chest radiograph.[28] Investigation for specific pathogens should be performed based on clinical and epidemiologic data if it would significantly alter management from standard empiric treatment. If the patient presents with gastrointestinal symptoms such as nausea, vomiting, or diarrhea in addition to the classic symptoms of pneumonia, *Legionella* is the likely culprit. A blood culture, sputum culture, and urine antigens for *S. pneumoniae or Legionella* may be performed.

Treatment

The first consideration in treating acute pneumonia is whether the patient requires inpatient or outpatient management. To guide this decision, two severity indices are commonly used in combination with psychosocial factors: (1) the pneumonia severity index, based on patient demographics, comorbidities, examination, and laboratory or radiologic findings,[5] and (2) the CURB-65 (*C*onfusion, *U*remia [>20 mg/dL], *R*espiratory rate [>30 breaths/min], low *B*lood pressure [<90/60 mm Hg], age *65* years or older).[28] Each factor is assigned a point. Recommendations for scoring include the following:

- 0–1: Outpatient treatment
- 2–3: Consider hospitalization versus outpatient treatment with close observation
- 3–5: Inpatient treatment (consider intensive care unit admission on an individual basis)

The Infectious Disease Society of America has published guidelines for empiric antibiotic therapy of community-acquired pneumonia.[28] For previously healthy patients treated as outpatients, macrolides are the first-line treatment. Patients with comorbidities, immunosuppression, or use of antimicrobial agents within the previous 3 months should be treated with a respiratory fluoroquinolone or a β-lactam agent plus a macrolide drug. With regard to duration of therapy, patients should be treated for a minimum of 5 days, be afebrile for 48 to 72 hours, and have stable vital signs.[28] A short-term inhaled bronchodilator may be used if the course is complicated by reactive airway disease.[27]

Return-to-Play Guidelines

As with other respiratory tract infections, athletes can gradually resume activity as long as they are afebrile, lack other systemic symptoms, and pass the "neck check."

BLOOD-BORNE INFECTIONS

It is important to understand and communicate the risk that blood-borne pathogens pose to athletes, how they are transmitted, and how they can be avoided. This section focuses on hepatitis B virus (HBV), hepatitis C virus (HCV), and HIV. Rare reports have been made of sports-related transmission, but it must be emphasized that the vast majority of infections are acquired from off-field behavior. High-risk behaviors include unsafe sexual practices and use of contaminated needles for blood doping and illicit injectable recreational and performance-enhancing drugs. Athletes at greatest risk for sports-related transmission are those who sustain bleeding injuries, have close contact with other players, and play in high-prevalence regions.

It is not recommended that athletes undergo mandatory testing, disclosure, or removal based on their blood-borne pathogen-infected status.[29] The following prevention strategies, published by the American Academy of Pediatrics, should be used to minimize the risk of transmission of blood-borne infections in the athletic setting[30]:

- Adhere to universal precautions when managing bleeding injuries: wear gloves when coming into contact with bodily fluids, remove and dispose of gloves immediately in appropriate containers, and then wash hands thoroughly with soap and water.
- Vaccinate all athletes and support staff coming into contact with injured athletes against HBV.
- Properly cover any wounds or damaged skin.
- Practice good personal hygiene and avoid sharing personal items such as razors, toothbrushes, and nail clippers that may be contaminated with blood.
- Educate athletes regarding the risks of acquiring blood-borne infections on and off the field.

Hepatitis B Virus

Among the blood-borne pathogens, HBV poses the greatest risk of transmission in the athletic setting. HBV is thought to be 100 times more transmissible than HIV and 10 times more transmissible than HCV.[30] HBV infections may clear spontaneously, but in some persons chronic infections develop, and they often have high viral loads. Fortunately a safe and effective HBV vaccine is available, and it is recommended that all athletes and support staff coming into contact with injured athletes be vaccinated.

HBV is highly prevalent in some regions of the world where perinatal transmission is common. The most efficient modes of transmission are through sexual intercourse and parenteral exposure. Mucosal contact with infected blood products can also occur, but to a lesser degree. Sports-related transmission has been reported in high-prevalence regions of the world when chronic carriers engaged in high-contact activities with uncovered wounds. Various professional athletic organizations publish specific guidelines regarding prevention strategies. The NCAA recommends considering the removal of athletes with acute HBV from participation in close-contact sports until they are negative for hepatitis B surface antigen and removal of chronic carriers from close-contact sports.[31] The International Olympic Committee, on the other hand, does not recommend barring athletes based on blood-borne pathogen-infected status.[30]

Hepatitis C Virus

Approximately 1.6% of the US population is infected with HCV.[30] Transmission is predominantly parenteral; most persons acquire the infection through intravenous drug injection. Perinatal and sexual transmission may also occur, but to a much lesser degree. No documented cases of sports-related transmission have been reported, but transmission has been reported in athletes during use of performance-enhancing drugs.[30]

Human Immunodeficiency Virus

The risk of transmission of HIV during athletic activity appears to be extremely low. To date, sports-related HIV transmission has not been reported. Evidence from health care workers suggests that the risk from mucocutaneous exposure is approximately 0.1%. The greatest risks to athletes remain high-risk off-field sexual behaviors and intravenous drug use.[30]

WATERBORNE DISEASES AND RECREATIONAL WATER-RELATED ILLNESS

Waterborne diseases can be transmitted through contaminated drinking water or participation in water-based sports in recreational water facilities. Participation in water-based sports and recreational activities presents a unique set of infectious disease concerns. Epidemiologically, swimmers have higher rates of illness than do nonswimmers, especially gastrointestinal, respiratory, eye-ear-nose, and skin symptoms.[32] Recreational water-related illnesses (RWIs) are associated with swimming in contaminated water venues including swimming pools, hot tubs, rivers, lakes, and oceans. Although most RWIs are self-limited, some infections may be life-threatening or progress to serious chronic conditions.

RWIs peak in the summer months, when recreational water facilities are most heavily used. The majority of waterborne disease outbreaks occur in treated water venues, resulting in 90.7% of cases of waterborne illness.[33] Inadequate treatment, bad hygiene, or equipment failures are responsible for most outbreaks. Higher rates of infection occur among children than among adults because of a naive immune system and behavioral factors such as poor hygiene and ingestion of recreational water. Infections are also more likely to be severe in children, who are more prone to dehydration because of their small size.

The spectrum of illnesses transmitted through recreational swimming from most to least common includes gastrointestinal (48%); skin (21%); respiratory (11%); and ear, eye, neurologic, and wound infections.[33] Diarrheal illnesses are the most commonly reported RWIs, caused by *Cryptosporidium, Giardia, Shigella, Escherichia coli* O157:H7, and norovirus. In treated water venues, *Cryptosporidium* is responsible for 66% of outbreaks. *E. coli* O157:H7 and *Shigella* spp. are responsible for 12% of outbreaks.[34] In freshwater venues, *E. coli* O157:H7 and *Shigella* spp. are the most common pathogens.[34]

Many cases of RWIs are linked to fecal contamination of water. Nonfecal shedding in the form of vomit, mucus, saliva, and skin also occurs. Disease may also spread through contact with contaminated surfaces, biofilms, and free-living organisms in pools, natural spas, hot tubs, and their components—heating, ventilation, and air conditioning—as well as other wet surfaces in recreational facilities. In recreational water facilities, chlorination is the primary defense against infectious disease. In such disinfected settings, disease may result from (1) poor maintenance of disinfectant levels (<1 ppm chlorine, equipment failure, inadequate or improper use of hydrogen peroxide and/or ultraviolet light), (2) chlorine resistance in waterborne pathogens, and (3) failure to practice healthy swimming habits.[32] Although the majority of infections occur in treated water facilities, the increasing popularity of ecotourism and adventure sports has given rise to a number of rare, serious infections including leptospirosis, *Naegleria* infection, and schistosomiasis.

The following general preventive measures are recommended to reduce the risk of waterborne infections:

- Persons with diarrhea should refrain from swimming while symptomatic and for 2 weeks after cessation of diarrhea (especially if diagnosed with cryptosporidium or *Giardia*).[35]
- Shower with soap and water before and after swimming.
- Wash hands with soap and water for at least 20 seconds after using a toilet or after changing diapers and before eating or drinking.
- Avoid swallowing pool water.
- Avoid urinating in pools and prevent direct animal access to recreational waters when possible.
- Take care not to overload facilities; shallow, heavily populated water carries a higher risk of infection.
- Avoid sharing water bottles, ice buckets, and eating utensils.

Fecally-Derived Recreational Water–Related Illnesses

Fecal contamination of water can occur through (1) accidental fecal release by humans, (2) residual fecal material on swimmers' bodies, or (3) direct animal contamination in outdoor pools.[35] Poor personal hygiene is a significant concern because an average of 0.14 g of feces is found on the perianal surface. Children can have up to 10 g of feces on the perianal surface.[34]

Cryptosporidiosis

Definition and Epidemiology

Cryptosporidium is a parasite that causes the diarrheal disease cryptosporidiosis. "Crypto" is a common term referring to both the parasite and the disease. *Cryptosporidium,* transmitted through both drinking and recreational water, is one of the most frequent causes of waterborne illness among humans in the United States. It is responsible for 66% of outbreaks in treated water facilities.[34] *Cryptosporidium* parasites are found worldwide and in every region of the United States. An estimated 748,000 cases of cryptosporidiosis occur in the United States annually.[35]

Pathobiology

Parasites are shed in the stool of persons infected with *Cryptosporidium*. Transmission is via the fecal-oral route and may occur if a person swallows recreational water contaminated with *Cryptosporidium* or ingests food or water that has come in contact with infected stool or contaminated surfaces. Numerous factors contribute to the high transmissibility of this parasite in recreational water facilities. The oocyst form of *Cryptosporidium* is highly resistant to environmental stress and disinfectants. The levels of chlorine required to kill *Cryptosporidium* oocysts are in excess of 30 mg/L, a level prohibitive to swimming. At standard chlorine levels, disinfection requires additional methods such as ozonation, ultraviolet radiation, or filtration. Infected hosts shed a high concentration of parasites, around 10^6 oocysts per gram of stool, and the infectious dose is relatively low; 132 oocysts infects 50% of persons.[32] Hosts may continue to shed oocysts for weeks after cessation of diarrhea.[32] After human ingestion of the oocysts, a 4- to 9-day incubation period occurs, after which the host experiences prolonged diarrhea, vomiting, and abdominal cramps. Gastrointestinal symptoms are usually self-limiting, resolving after 1 to 3 weeks. Some persons may experience a recurrence of symptoms after a brief period of recovery, and symptoms can come and go for up to 30 days.[35] Persons with weakened immune systems, such as those with HIV or those taking immunosuppressive medications, may experience chronic, severe, life-threatening illness.

Diagnosis

The differential diagnosis includes other parasitic, viral, and bacterial gastrointestinal infections. Cryptosporidiosis should be suspected especially if the diarrhea is prolonged, lasting longer than 48 hours. Diagnosis may be made by microscopically visualizing oocysts in the stool using acid-fast staining, direct fluorescent antibody, and/or enzyme immunoassays for *Cryptosporidium* spp. antigens. Detection may be difficult and require several stool samples. Molecular methods such as the polymerase chain reaction (PCR) also may be used.[35]

Treatment

For most persons with healthy immune systems, cryptosporidiosis is a self-limited, self-resolving illness. Preventing dehydration by drinking plenty of fluids is the most important management principle. Young children and pregnant women are especially susceptible to dehydration. Nitazoxanide, 500 mg by mouth twice daily for 3 days, is approved by the US Food and Drug Administration for treatment of diarrhea caused by *Cryptosporidium* in patients with healthy immune systems age 12 years and older, but it may have limited efficacy in immunocompromised persons.[35]

Return-to-Play Guidelines

Persons should refrain from swimming while they are experiencing diarrhea. Those diagnosed with cryptosporidiosis should avoid, swimming for at least 2 weeks after the diarrhea stops. Good personal hygiene should always be practiced: washing hands before preparing or eating food, showering before entering the water, and avoiding swallowing of recreational water.[35]

Infectious Diarrhea

Cryptosporidium is the most common cause of infectious diarrhea acquired through recreational water sports because of its unique resistance to chlorine. Gastrointestinal infections constitute the vast majority of RWIs and may be caused by a number of parasites, bacteria, and viruses. Other causes of infectious diarrhea—which may be transmitted through recreational water, drinking water, or food—are discussed here.

Epidemiology and Pathobiology

High school and college athletes are at risk for gastrointestinal infections because of exposure to common sources of food, water, the environment, person-to-person contact, and the dormitory lifestyle, in which students share bathrooms, living areas, and snack foods. Swimming is an independent risk factor for *Giardia* infection. *Giardia,* like *Cryptosporidium,* is a parasite with a cyst form that is resistant to harsh environmental conditions. However, *Giardia* is much more sensitive to chlorine and is inactivated by standard disinfectant levels. Giardiasis presents with diarrhea, cramps, loss of appetite, fatigue, and vomiting usually lasting 7 to 10 days.[36]

The most common causes of bacterial infectious diarrhea are *E. coli* O157:H7 and *Shigella* spp. Both bacteria are susceptible to chlorine, but failure to maintain disinfection levels may result in infection. *E. coli* O157:H7 initially presents as a nonbloody diarrhea but may have serious complications. Infection can progress to hemorrhagic colitis, and 5% to 10% of persons infected with *E. coli* experience hemolytic uremic syndrome. Patients with hemolytic uremic syndrome present with hemolytic anemia and acute renal failure, often accompanied by vomiting and fever. *E. coli* diarrhea usually resolves in 1 week but the organism may be shed in the stool for 7 to 13 days. *Shigella* infections present with diarrhea, fever, and nausea. *Shigella* is typically a self-limiting infection lasting 4 to 7 days, but the organism may be shed in the stool for 30 days.[36]

Viruses cannot multiply in water, but they may be present in recreational water as a result of shedding by a person during and after infection. In the past 2 decades, an increase in viral outbreaks has occurred, with a preponderance occurring in treated water venues. Sinclair and colleagues[32] have reported that among all known waterborne disease outbreaks between 1951 and 2006, 49% occurred in treated water and 40% occurred in freshwater venues such as lakes and rivers. Inadequate disinfection was linked to 69% of cases involving treated water venues. Six types of viruses are shed after an infection: rotavirus, norovirus, adenovirus, astrovirus, enterovirus, and hepatitis A virus.[32] The most common viral pathogens are noroviruses (45%), adenovirus (24%), echovirus (18%), hepatitis A virus (7%), and Coxsackieviruses (5%).[32] Norovirus is the most common cause of gastroenteritis in adults. Within 48 hours of exposure, patients present with 24 hours of diarrhea, nausea, vomiting, and fever. Viral shedding can occur for up to 5 days after the onset of symptoms. Norovirus may be somewhat chlorine-resistant. Many fecally-derived viruses also cause extraintestinal symptoms; these viruses are discussed separately.

Diagnosis

Most cases of infectious diarrhea are self-limiting, and thus laboratory tests are not routinely needed. In severe or recalcitrant cases, multiple diagnostic tests are available to identify the infectious pathogen. These tests include but are not limited to fecal leukocyte determination, stool culture for enteric pathogens and parasites, stool PCR, *Clostridium difficile* toxin testing, and endoscopy. Laboratory testing may be indicated if the patient has grossly bloody stools, profuse diarrhea with dehydration, duration of illness longer than 48 hours, fever, severe abdominal pain, or diarrhea in immune-compromised or elderly persons.

Treatment

The mainstay of therapy for most cases of infectious diarrhea is supportive treatment and adequate hydration. Treatment of more severe or prolonged cases of infectious diarrhea should be guided by laboratory testing and diagnosis of the specific causative agent.

Return-to-Play Guidelines

Persons should refrain from swimming while experiencing diarrhea and generally for 2 weeks after cessation of symptoms.

Other Fecally-Derived Waterborne Diseases

1. *Adenovirus* may be transmitted both fecally and nonfecally and can present with gastrointestinal infection or pharyngoconjunctival fever.
2. *Hepatitis A* is transmitted through the fecal-oral route and presents with anorexia, nausea, vomiting, and jaundice after an incubation of 15 to 50 days. Infected persons may shed virus prior to the onset of symptoms.
3. *Aseptic meningitis* caused by enteroviruses has occurred in the high school football setting and during an international

soccer tournament in Belgium.[37] The most common pathogens are echoviruses (5, 9, 16, and 24) and Coxsackie viruses (B1, B2, B4, and B5). Infections peak during the summer months. Transmission through sharing of water bottles or drinking cups, ice buckets, and water coolers have been responsible for outbreaks in the athletic setting. Athletes should be discouraged from sharing drinking vessels and ice buckets and encouraged to wash their hands frequently. Echovirus meningitis has also been associated with recreational water use.[32]

NONFECALLY-DERIVED WATERBORNE DISEASES

Pseudomonas Infections

Pseudomonas infections acquired in the setting of recreational water sports have two main clinical manifestations: (1) acute otitis externa (swimmer's ear), commonly acquired in swimming pools, and (2) hot tub folliculitis, acquired in hot tubs and spas.

Pathobiology

Pseudomonas is ubiquitous in water, vegetation, and soil, with a preference for warm, moist environments. *Pseudomonas* bacteria are shed by bathers in pools and hot tubs and on surrounding wet surfaces; they can also accumulate in biofilms in filters. In hot tubs, high temperatures and turbulence promote perspiration and desquamation or shedding of epithelial cells. The increased organic load in the water both provides nutrients for bacterial growth and decreases residual disinfectant levels. *Pseudomonas* is chlorine-sensitive; thus all outbreaks have occurred as a result of inadequate disinfection. Patient risk factors for infection include communal use of an athletic spa, increased duration of exposure, and the presence of skin trauma, such as turf burns and abrasion due to body shaving.[36]

Hot Tub Folliculitis

Hot tub folliculitis is an infection of hair follicles caused by the bacterium *P. aeruginosa.* When bathers use a hot tub, the warm water supersaturates the epidermis, dilating dermal pores and facilitating bacterial invasion. Extracellular enzymes produced by *Pseudomonas* damage the skin. A pustular rash appears anywhere between 8 hours to 5 days after exposure and usually resolves spontaneously within 5 days. The rash is generally more severe in areas that bathing suits cover because swimwear traps the bacteria. Some people report other symptoms such as headache, muscle ache, burning eyes, and fever.[12] This self-limiting infection generally does not require treatment.

Acute Otitis Externa

Definition and Epidemiology

Acute otitis externa (AOE), also known as "swimmer's ear" or "tropical ear," is defined as generalized inflammation of the external ear canal, which may also involve the pinna or tympanic membrane. AOE is a very common infection. The annual incidence of AOE is between 1:100 and 1:250 of the general population, and the lifetime incidence is up to 10%.[38] AOE is more common in regions with warmer climates and increased humidity and with increased water exposure from swimming. Patient risk factors are increased amounts of time spent in water, age less than 19 years, and a history of previous ear infections.

Pathobiology

In North America, nearly all cases of AOE (98%) are caused by bacteria. The most common pathogens are *P. aeruginosa* (20% to 60%) and *S. aureus* (10% to 70%). Gram-negative bacteria other than *P. aeruginosa* are implicated in 2% to 3%, and polymicrobial infections are common. Fungal involvement is uncommon in primary AOE but may be found in persons with chronic otitis externa or after treatment of AOE with topical antimicrobial agents.[38] Cerumen, with a slightly acidic pH, helps to inhibit infection. Repeated water exposure removes the protective wax coating of the external ear canal. Other behaviors that alter this natural barrier—including removal of cerumen by cleaning of the ear canal, leaving soapy deposits, or use of alkaline eardrops—may increase a person's risk for AOE. Other factors are local trauma, sweating, allergy, and stress.

Diagnosis

The classic presentation of diffuse AOE is tenderness of the tragus (when pushed) and/or pinna (when pulled up and back) that is intense and disproportionate to what might be expected based on visual appearance. Diagnosis is made on the basis of the following clinical criteria[38]:

1. Rapid onset, within 48 hours, in the past 3 weeks AND
2. Symptoms of ear canal inflammation including otalgia, itching, or fullness WITH OR WITHOUT hearing loss or jaw pain AND
3. Signs of ear canal inflammation that include tenderness of the tragus, pinna, or both OR diffuse ear canal edema, erythema, or both, WITH OR WITHOUT otorrhea, regional lymphadenitis, tympanic membrane erythema, or cellulitis of the pinna and adjacent skin.

Alternative diagnoses to consider are acute otitis media, furunculosis (localized AOE usually caused by *S. aureus*), otomycosis, herpes zoster oticus, and contact dermatitis. Unlike the case with acute otitis media, the tympanic membrane should demonstrate normal tympanic mobility in persons with AOE.

Treatment

The clinical practice guidelines released by the American Academy of Otolaryngology–Head and Neck Surgery Foundation in 2006[38] include the following key points:

1. Assess for pain and recommend appropriate analgesic treatment.
2. Use topical preparations for initial therapy of diffuse, uncomplicated AOE. Appropriate topical treatments include acetic acid, boric acid, aluminum acetate, silver nitrate, and *N*-chlorotaurine. A topical steroid is effective as a single agent or in combination with acetic acid. Symptoms should improve or resolve in 48 to 72 hours.
3. Systemic antimicrobial therapy should be used only if there is extension outside of the ear canal or for specific host factors such as diabetes, an immunocompromised state, or a history of radiotherapy.

Prevention strategies attempt to limit water accumulation and retention in the external auditory canal and to maintain a healthy skin barrier. The following measures may help to reduce the risk of AOE: (1) remove obstructing cerumen; (2) use acidifying ear drops shortly before swimming, after swimming, or at bedtime; (3) use a hair dryer to dry the ear canal; (4) use ear plugs while swimming or showering; and (5) avoidance of trauma to the external auditory canal.[38]

Return-to-Play Guidelines

It is recommended that patients with AOE refrain from water sports for 7 to 10 days during treatment. Competitive swimmers may sometimes return to competition with well-fitted earplugs 2 to 3 days after treatment or after the pain has resolved.

NEUROLOGIC INFECTIONS

Meningitis and Encephalitis

Definition and Classification

Acute meningitis is a medical emergency that requires prompt recognition and treatment. Delay in treatment can result in significant morbidity or death. Meningitis is defined as inflammation of the meninges—that is, the membranes lining the brain and spinal cord. Encephalitis is inflammation of the brain parenchyma. The term "meningoencephalitis" refers to a combination of these two processes and the disease usually has a viral etiology.

Meningitis is often classified as either septic (bacterial) meningitis or aseptic (usually viral) meningitis. Septic meningitis is most commonly caused by associated *N. meningitides* (meningococcus) and *S. pneumoniae* (pneumococcus). Aseptic meningitis, usually of viral etiology, is much more common and refers to a clinical syndrome of meningitis with negative Gram stain and bacterial culture of the cerebrospinal fluid (CSF). Meningitis is also classified according to its disease course. Acute meningitis presents within 1 day of onset of symptoms, subacute meningitis occurs within 1 to 7 days of onset, and chronic meningitis persists for 4 or more weeks.

Epidemiology

N. meningitidis outbreaks have been reported among soccer and rugby players.[2] Additionally, multiple reported outbreaks of acute viral meningitis occurred in high school football teams in the United States between 1960 and 1993. This population may be at elevated risk over the general population for several reasons, including repeated close physical contact, close living quarters, and the practice of sharing water bottles. Enteroviruses are usually transmitted via the fecal-oral route but may also be transmitted via respiratory secretions. Finally, the US football season coincides with the peak enteroviral season (late summer and autumn).[39]

Pathobiology

Acute aseptic meningitis is the most common form of meningitis and is usually caused by viruses. Other causes include Lyme disease, tuberculosis, and rickettsia.[40] The most common causes of septic meningitis are *S. pneumoniae* and *N. meningitidis*. *Listeria monocytogenes* and group B streptococcus also cause septic meningitis. Cases of *H. influenza* meningitis have declined considerably since the initiation of universal vaccination.

Diagnosis

Management of a patient with suspected meningitis is driven by a critical fact—that a delay in the initiation of therapy significantly increases the already high risk of morbidity and mortality. The practice guidelines published by the Infectious Diseases Society of America present an algorithm for pursuing diagnostic testing in a manner that does not compromise the timely delivery of antimicrobial and adjuvant therapy.[41]

The hallmark symptoms of meningitis are fever, headache, and meningismus (stiff neck). Patients may also have nausea, vomiting, photophobia, malaise, and drowsiness. Altered consciousness is highly suggestive of encephalitis. Septic meningitis is a medical emergency that usually presents with more severe symptoms than aseptic meningitis but may be clinically indistinguishable. All cases of suspected meningitis should be presumed septic until proven otherwise. Initial assessment should determine airway status, respiratory effort and rate, and circulation.

The physical examination should focus on the following goals: (1) detecting signs to support the diagnosis of meningitis, (2) signs that may help identify the offending pathogen, and (3) signs of cerebral edema prior to performing a lumbar puncture. The examiner should check for other signs that may explain symptoms, such as trauma, conjunctivitis, intraocular hemorrhage, or papilledema. The patient is then evaluated for nuchal rigidity, or a decrease in neck suppleness, a sign of meningeal irritation due to infection or hemorrhage. With the patient relaxed, the examiner should first move the head in all planes to assess for resistance to motion. While the patient is supine and passively flexing the neck forward, the examiner should observe the hips and knees. Two signs of meningeal irritation are as follows: (1) flexion of the hips and knees with neck flexion represents a positive Brudzinski sign and (2) pain or resistance to flexion of the hip to 90 degrees with the knee extended represents a positive Kernig sign. A bilateral positive Kernig sign may distinguish meningeal irritation from lumbrosacral nerve root compression, which is usually unilateral. A complete neurologic examination should be performed, including cranial nerve, strength, sensation, and cerebellar testing. Focal neurologic deficits are unusual for most types of acute meningitis and may suggest a mass effect or an atypical infection.

For clues regarding the specific etiology of meningitis, the examiner should perform a skin examination with full exposure. Hemorrhagic skin lesions such as petechiae or purpura are associated with meningococcal meningitis, especially noted most commonly on the limbs. A rash in a stocking-glove distribution is suggestive of a rickettsial infection. Herpetic lesions may implicate HSV.[39]

Lumbar puncture and analysis of the CSF is important for diagnosing meningitis. Although it may reduce the sensitivity of CSF analysis, empiric antibiotic therapy should not be delayed while attempting to perform a lumbar puncture or other testing. Cerebral herniation is a feared complication of lumbar puncture. If the patient is immunocompromised and has a history of CNS disease, papilledema, or focal neurologic deficits, blood cultures should be drawn, empiric antimicrobial therapy should be initiated, and a CT scan should be obtained before a lumbar puncture is performed.[41] Gram stain and culture of CSF should be obtained, but their sensitivity may be reduced by antimicrobial therapy. In persons with aseptic meningitis, the CSF classically demonstrates mononuclear pleocytosis with less than 1000 white blood cells/mm^3 and a normal or mildly decreased glucose level. Persons with septic meningitis have a neutrophil-predominant pleocytosis with more than 1000 white blood cells, a low glucose level, and a high protein level. PCR of CSF can detect enterovirus or HSV. Enterovirus can also be detected in the stool or oral mucous membranes.

Treatment and Prevention

For persons with acute meningitis, the primary management goal is early empiric antibiotic treatment in an inpatient setting, preferably with ready access to intensive care. Empiric antimicrobial therapy should be initiated once bacterial meningitis is diagnosed by CSF analysis or upon high clinical suspicion if there is a delay in lumbar puncture.[41] For adults and children between the ages of 2 to 50 years, empiric therapy includes vancomycin plus a third-generation cephalosporin.[41] The spectrum of antibiotic coverage should be narrowed as soon as possible on the basis of CSF culture and susceptibility results. Empiric treatment with intravenous acyclovir is also recommended if the clinical picture suggests HSV infection. Adjunctive dexamethasone is recommended for adults with pneumococcal meningitis.[41]

Routine vaccination with meningococcal vaccine is recommended for college freshmen living in dormitories and other populations at increased risk.[3] For secondary prevention, persons with "high risk" exposure after a confirmed case of meningococcal meningitis should receive chemoprophylaxis with ceftriaxone, rifampin, ciprofloxacin, or passive immunization. The American Academy of Pediatrics Infectious Disease Committee defines "high risk" contacts as household contacts, child care contacts, and anyone with direct exposure to the index patient's secretions through kissing or sharing toothbrushes or eating utensils during the 7 days before the onset of illness.[39]

Return-to-Play Guidelines

Infected players should be isolated from the rest of the team as soon as possible. To date, no evidence-based guidelines exist for RTP from meningitis. As with any other infection, the athlete should be afebrile and have no constitutional signs, and it would be prudent to refrain from play until all symptoms have resolved. A careful neurologic examination should be performed, including a hearing test, and any changes should be noted before RTP. Viral shedding may occur for several weeks after resolution of symptoms; therefore athletes should be counseled on adequate hand hygiene, advised to avoid sharing ice buckets or drinking vessels, and dissuaded from refilling receptacles by dipping them in team coolers. School and local public health authorities should be informed when an outbreak occurs on an athletic team to address risk to close contacts and the community at large.

UNUSUAL INFECTIONS

The increasing popularity of adventure sports and ecotourism, such as triathlons and kayaking, have fostered some rare yet potentially life-threatening infections that are discussed in this section.

Leptospirosis

Definition and Epidemiology

The bacterium *Leptospira interrogans* causes the disease leptospirosis, also known as swineherd's disease, Stuttgart disease, and Weil syndrome. Several outbreaks of leptospirosis have occurred among triathletes and outdoor adventure participants since the 1980s.[42] Leptospirosis occurs worldwide but is more prevalent in temperate or tropical climates. Swimming, wading, kayaking, and rafting in freshwater contaminated with animal urine are associated with leptospirosis.

Pathobiology

Many animals carry *L. interrogans,* including cattle, pigs, horses, dogs, and rodents. Humans can become infected via direct contact with urine from infected animals or contact with water, soil, or food contaminated with the urine of infected animals. Bacteria enter the body through skin cuts and abrasions as well as mucosal surfaces of the mouth, nose, and conjunctiva. Measures should be taken to prevent direct animal access to recreational water. The majority of cases occur in bodies of freshwater such as rivers and lakes or unchlorinated swimming pools. However, *L. interrogans* is susceptible to standard disinfection and has a low resistance to harsh physical conditions.[36] Upon exposure to the bacteria, there is an incubation period of 2 days to 4 weeks. Illness usually begins abruptly with fever, chills, headache, muscle aches, vomiting, or diarrhea. After a period of recovery, a second phase of illness may occur in some patients. The second phase, also called Weil disease, is a severe and potentially fatal condition associated with liver and kidney failure, hemorrhagic jaundice, and aseptic meningitis.[43]

Diagnosis and Treatment

The most sensitive laboratory test early in infection is an IgM enzyme-linked immunosorbent assay for *Leptospira* antibodies. Leptospirosis is treated with doxycycline. More severe infections may require administration of intravenous antibiotics. Persons traveling to areas where leptospirosis is endemic or epidemic and participating in high-risk exposure activities such as swimming or kayaking in freshwater may be benefit from preexposure chemoprophylaxis. Several studies have demonstrated the efficacy of preexposure oral doxycycline 200 mg orally, weekly, begun 1–2 days before and continuing through the period of exposure in reducing clinical symptoms and mortality rate.[43]

Naegleria fowleri

Definition and Epidemiology

Naegleria fowleri, a free-living amoeba that causes a very rare but usually fatal illness in humans called primary amebic meningoencephalitis, is found around the world in soil and bodies of warm freshwater such as lakes, rivers, and hot springs. *N. fowleri* prefers warm water and can reproduce in temperatures up to 46°C.[36] Most infections in the United States have occurred in the southern states. From 2001–2010, a total of 32 infections were reported in the United States; 30 people were infected by contaminated recreational water and 2 were infected from a geothermal drinking water source.[44]

Pathobiology

Infection occurs through forceful inhalation or splashing of contaminated water into the olfactory epithelium, which can occur through diving, jumping, underwater swimming, or nasal irrigation. The amoebae then move into the brain and CNS. After an incubation period of 7 to 10 days, the patient presents with symptoms and signs resembling bacterial meningitis, including headache, high fever, stiff neck, nausea, vomiting, seizures, and hallucinations. Death usually occurs 3 to 10 days after the onset of symptoms.[12]

Treatment

No known treatment exists for *N. fowleri*, and almost all infections are fatal. The only definite way to prevent infection is to refrain from water-related activities in warm, untreated, or poorly treated water. Measures that may reduce risk of infection include the following:

1. Avoidance of water-related activities in warm freshwater during periods of high water temperature and low water levels
2. Holding the nose shut or using nose clips during water-related activities
3. Avoidance of digging or stirring the sediment while swimming in warm freshwater areas
4. Using only water that has been distilled, sterilized, or boiled to irrigate the sinuses

Schistosomiasis

Definition and Epidemiology

Schistosomiasis, a parasitic disease caused by *Schistosoma haematobium, S. japonicum, and S. mansoni,* affects approximately 300 million people worldwide. The parasite lives in freshwater in most African countries and some parts of South America, the Caribbean, the Middle East, and Asia. The highest prevalence and parasite load occurs in children aged 5 to 15 years. The emergence of ecotourism and adventure tourism has resulted in a number of imported cases since the 1980s.[45]

Pathobiology

Eggs are excreted in the feces or urine of infected hosts. Snails act as intermediate hosts, releasing cercariae, the infective form. Humans are infected by swimming in contaminated freshwater, when cercariae swim and penetrate the skin. In the human host, the parasite migrates through several tissues and stages before settling in the mesenteric venules. *S. japonicum* and *S. mansoni* usually occupy the superior mesenteric veins draining the small and large intestines, causing gastrointestinal and hepatosplenic schistosomiasis, whereas *S. haematobium* often resides in the venous plexus of the bladder, causing urinary schistosomiasis.[46]

Most infections are asymptomatic and often overlooked. Contact with the penetrating cercariae may cause "swimmer's itch" or "cercarial dermatitis," a micropapular dermatitis that disappears within 48 hours after contact with contaminated water. Katayama fever may occur 3 to 6 weeks after heavy exposure and is an acute toxemic syndrome that affects a varying percentage of exposed persons. Hallmark clinical features include fever, malaise, and eosinophilia. Less commonly hepatosplenomegaly, diarrhea, urticaria, and edema can occur.[45]

Left untreated, transient symptoms disappear and infection progresses to a chronic phase. *S. haematobium* typically causes urinary schistosomiasis, with hematuria, proteinuria, leukocyturia, dysuria, and nocturia. Patients with chronic urinary schistosomiasis experience obstructive uropathy, hydronephrosis, and calcified fibrotic bladder and/or ureter. *S. mansoni, S. japonicum, S. mekongi, and S. malayi* typically cause gastrointestinal and hepatosplenic schistosomiasis. Patients initially present with bloody diarrhea, followed by the development of chronic colitis with possible colonic polyps. Severe, long-standing infections may lead to portal hypertension caused by fibrosis induced by retained eggs. CNS involvement may occur after embolization of eggs from the portal mesenteric system.[45]

Diagnosis

Diagnosis is usually made on clinical and epidemiologic grounds. Eggs may not be detectable in the stool during the acute phase because of a low parasitic load. Serologic confirmation of the diagnosis based on antigen detection is possible 5 to 6 weeks after the acute infection. Tissue biopsy or ultrasonography may also aid in diagnosis.

Treatment

Praziquantel at a dose of 40 mg/kg per day orally in two divided doses for 1 day is effective against all *Schistosoma* spp., but it kills only the adult worm. *S. japonicum* requires a higher dose of 60 mg/kg per day orally in three divided doses for 1 day. If treatment is started during the initial phase of illness, a second dose is required once all worms have reached the adult stage. The drugs trichlorfon and oxamniquine are also used in areas where a single species exists. Corticosteroids are helpful in reducing the immunologic reaction in Katayama fever, but with the use of corticosteroids, praziquantel dosing must to be increased to 20 mg/kg every 12 hours for 3 days.

For a complete list of references, go to ExpertConsult.com.

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Citation:
Turbeville SD, Cowan LD, Greenfield RA. Infectious disease outbreaks in competitive sports. *Am J Sports Med*. 2006;34:1860–1865.

Level of Evidence:
IV

Summary:
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Sedgwick PE, Dexter WW, Smith CT. Bacterial dermatoses in sports. *Clin Sports Med*. 2007;26:383–396.

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IV

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Citation:
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Level of Evidence:
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